

Conceptual Questions and Answers: Order Flow Imbalances (OFI)

1. What’s the motivation behind measuring OFI at multiple depth levels of the order book?

The reward is that price stress and liquidity aren’t just visible on the top bid and offer — deeper levels in the order book contain important information about trader intentions and ensuing price movement. Orders get placed or canceled at multiple levels, especially in high-volatility or low-volatility stocks. These actions are affecting market forces deeper than what appears on the surface.

In Section 2.1.2 of the paper, authors define ”Deeper-Level OFI” and illustrate that multi-level OFIs are highly correlated, in which the first principal component explains over 89% of variance (see Table 2 and Figure 1). That is, employing multiple depth levels within OFI captures much richer order book dynamics, which helps in more precise return prediction and avoids overreliance on merely the most obvious orders.

2. Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

If you model the impact of other stocks’ OFIs on a specific stock’s return (cross-impact), you may have dozens of potential predictors (e.g., OFIs of 100 stocks) and relatively few observations, especially with high-frequency intraday data. In such an environment, Ordinary Least Squares (OLS) does not work well as it is prone to overfitting and instability.

Instead, they use Lasso regression, a combination of regularization and variable selection. It eliminates irrelevant predictors and shrinks the rest, leading to a sparser, more interpretable model with improved generalization. They discuss this in Section 3.1.5, where they justify Lasso specifically since ”the number of variables exceeds the number of observations” and there is high multicollinearity among OFIs of different assets.

3. Why is OFI considered a better predictor of short-term returns than trade volume?

Trade volume will inform you of the level of trading that has occurred but will not tell you who was more aggressive, the seller or buyer. But OFI directly measures imbalances in terms of buying and selling pressure by looking at how quotes and sizes on order books are changing. That also implies whether prices in bids are rising or falling, whether orders are being added or removed, etc.

As discussed in Section 2.1.1, and in accordance with earlier work the paper references (e.g., Cont et al., 2014), short-horizon price action is determined mainly by OFI, instead of raw trade volume. Experiments in the paper confirm that models based on OFI systematically outperform volume or history-based models as well as return-based models in terms of explanatory power and out-of-sample forecasting (see Table 8 and Section 4.2.1).